📱 MOBILE PHONE PRICE PREDICTION

Feature engineering, exploratory analysis and machine-learning valuation of Indian retail handset listings

Prepared by **Srishti Sharma**

Dataset Processed_Flipdata.xlsx · 531 listings × 12 raw columns · 187 unique models

531 rows Final rows	**23** Final analysis columns	**10** Duplicate listings removed	**2.8%** Flagged records	**187** Unique models
0.751 Held-out R^2	**0.688** Grouped-CV mean R^2	**0.582–0.769** CV range	**₹3,206** Test MAE	₹7,040 Naive MAE

1. Executive Summary

This report develops a predictive model for mobile handset pricing from device specifications, using a 531-listing dataset of Indian retail phone listings after de-duplication. Unlike a formula-derived target, price here is market-driven: the analysis establishes which specifications are associated with price, and by how much, rather than confirming a known mathematical relationship.

The dataset required substantive remediation before analysis. Three defects were material: a mislabelled screen-size column storing values in inconsistent units, text-encoded camera specifications, and a 15-listing cluster of older budget models carrying garbled entries. The cluster was traced, corrected and carried explicitly through every downstream stage
rather than deleted. A fourth item — drift in the target column name across export runs — was handled defensively at load time as a precaution rather than a defect in this export.

Model selection ran entirely on group-aware cross-validation. A tuned Random Forest (max_depth 15, min_samples_leaf 2, n_estimators 500) was retained, achieving a test R^2 of 0.7507 and mean absolute error of ₹3,206 against a naive benchmark of ₹7,040, on RMSE of ₹5,548 and MAPE of 19.67%. A separate stability check produced a mean cross-validated R^2 of 0.688 across a 0.582–0.769 range. Both figures are reported together, because the single test-split number sits near the favourable end of that range and should not be quoted alone.

Central finding

Price is substantially specification-driven below the flagship tier, where storage (Memory), RAM and camera specifications explain most of the variation and brand identity carries little independent pricing power once specifications are known. At the flagship tier a premium is associated with Apple and Google specifically — though this finding rests on very few listings per brand (approximately two to nine) and should be read as directional rather than precise. This two-regime structure is the central strategic implication of the analysis and is developed in Section 5.

2. Data, Quality Remediation and Feature Engineering

2.1 Dataset Overview

The source export comprised 541 raw listings across 12 columns, one of which was a residual index column carried over from a previous save. Dropping it left 11 substantive raw fields and exposed 10 exact duplicate listings; removing those left 531 records covering 187 unique handset models. No raw nulls were present; ambiguous 0MP camera entries were treated as missing rather than read as zero and imputed using training-derived brand medians. The modelling split is 428 training and 103 test rows, grouped on handset model with zero overlap.

Dataset composition and post-cleaning position

Metric	Value
Total Listings	541 raw → 531 after removing duplicates
Raw Columns → Engineered Columns	12 raw (11 after dropping the residual index column) → 23
Duplicate Listings	10 exact duplicates (first record retained)
Missing Values	0 raw nulls; ambiguous 0MP camera entries imputed on brand median
Flagged Data-Quality Cluster	15 listings (2.8%), retained and flagged
Price Range	₹920 – ₹80,999
Average / Median Price	₹16,306 / ₹13,999
Flagged-cluster average price	₹1,851 vs. ₹16,726 for the remainder
Unique Models	187
Train / test split	428 / 103 rows, grouped on Model — zero overlap

2.2 Data-quality defects identified and resolved

– **Mislabelled screen-size column.** The "Mobile Height" field stored a subset of rows in inches, mixed with centimetres.Values were traced and converted to a consistent centimetre scale rather than dropped or averaged away.

– **Text-encoded camera specifications.** "50MP" and "8MP" strings were cast to numeric. Ambiguous 0MP entries were treated as missing and imputed using training-derived brand medians, not read as a literal zero.

– **Flagged data-quality cluster.** Fifteen listings (2.8%) sharing a garbled unit pattern were retained and flagged rather than silently deleted. A permanent is_flagged_dq indicator carries the cluster's identity into the trained model as a genuine feature, and the cluster was traced through outlier detection, correlation and residual analysis.

– **Safeguard — target column name drift.** Source exports have used both "Price" and "Prize" spellings across runs; this is handled defensively at load time. A safeguard against a known export inconsistency rather than a defect in this file.

2.3 Feature engineering

Six derived features convert raw specifications into modelling inputs: Brand (first token of the model-name string, with typo and alias correction); Base_Colour (15 keyword-matched colour families, with buckets below 10 observations folded into "Other"); Processor_Brand (chipset text mapped to Qualcomm, MediaTek, Exynos, Unisoc, Google or Apple); Is_5G (a boolean parsed from the model-name string); Total_Camera_MP (rear and front megapixels combined into a single camera- capability signal); and is_flagged_dq. The final model feature set comprises six numeric and five categorical fields, expanding to 42 columns after one-hot encoding.

3. Key Statistical Findings

Univariate, bivariate and multivariate structure were examined in sequence, with effect sizes and significance testing applied throughout. Two findings emerged only after the flagged data-quality cluster was isolated, and are set out below alongside a signal that the same investigation correctly ruled out.

Finding	Detail
Strongest Price Drivers	Memory ($r = +0.564$), RAM ($r = +0.529$) and Front Camera MP ($r = +0.504$, non-linear). Memory and RAM exhibit a statistically confirmed sub-additive interaction (coefficient −0.0246, $p = 0.017$), and are themselves correlated at $r = +0.644$ — so RAM adds little marginal information once Memory is in the model.
Brand Vs. Chipset	Effectively tied ($\varepsilon^2 = 0.366$ vs. 0.368, Kruskal-Wallis) — largely the same signal, not two independent levers. Their categorical association is directionally strong (Cramér's V = 0.699), but the underlying contingency table has sparse cell counts; 12 of the 15 categorical pairs tested fail the reliability check. Treat as suggestive, not confirmed.

Hidden Relationship Uncovered	Battery capacity showed no relationship in the full dataset (r = −0.046) but a moderate negative association (r = −0.441) once the 15-listing cluster was excluded — a masking effect visible only after deliberate investigation, and treated as an association rather than a causal effect.
False Signal Correctly Ruled Out	Screen size appeared to predict price (r = +0.149) but reversed to no relationship (r = −0.037) once the same cluster was excluded — a statistical artefact, not a real pricing factor.
Front Vs. Rear Camera	Front-camera megapixels carry a price-per-MP estimate approximately 4.5× larger than the rear camera (₹620 vs. ₹139 per MP in the fitted model). Rear-camera MP shows weaker differentiation in a dataset where 257 of 531 listings (48%) carry a 50MP rear camera.

Pearson correlations on the full 531-listing dataset. Categorical effect sizes are ε^2 from Kruskal-Wallis, not η^2 from one-way ANOVA: both Brand and Processor_Brand fail Levene's test for equality of variance, so the ANOVA-based statistic is not valid here. All coefficients are associations, not established causal effects.

4. Model Performance

4.1 Benchmarking approach

Nine algorithms were compared under group-aware cross-validation (GroupKFold), which ensures that the same handset model never appears in both training and validation folds — a stricter, leakage-free comparison than a random split. Selection and hyperparameter tuning ran entirely on cross-validation; the held-out test set was touched once, for the top three cross-validated finalists, to confirm the choice rather than to make it. SVR ranked fourth after tuning (CV R^2 0.534) and therefore never touched the test set at all.

The value of this protocol is visible in the Decision Tree. It ranked first on a single random split (test R^2 0.745) but eighth of nine under grouped cross-validation (0.365) — a discrepancy large enough to show that a single split cannot carry a ranking at this sample size. The test set was subsequently reused for residual and permutation diagnostics and for evaluating the alternative techniques in Section 4.4, so those later test-based comparisons are directional rather than independent.

Model	Training-Subset Pooled OOF R^2	Test R^2	Test MAE	Status
Naive (Mean Predictor)	−0.028	-0.000	₹7,040	Floor
Linear Regression	0.374	0.511	₹3,499	Untuned Baseline
Lasso (Best Linear Model)	0.499	0.688	₹2,743	Best Interpretable Model
XGBoost (tuned)	0.612	0.644	₹3,337	Highest pooled-OOF R^2 on the training subset
Random Forest (Tuned)	0.590	0.751	₹3,206	★ Final Model

Gradient Boosting (Tuned)	0.595	0.723	₹2,968	Strong Runner-Up (Best MAE)

CV R^2 is each model's best five-fold GroupKFold score — untuned for Linear Regression and Lasso (both shown at their cross-validated α), tuned for the ensembles. Test figures come from the single 428 / 103 held-out split at seed 42. The two bases use different samples and are not directly comparable. Figures in parentheses denote negative values.

4.2 Stability and the honest headline

Five-fold grouped cross-validation for the retained Random Forest returned per-fold scores of 0.582, 0.710, 0.753, 0.769 and 0.624 — a mean of 0.688 with a standard deviation of 0.073. Five repeated random splits produced a mean R^2 of 0.584 with a standard deviation of 0.118 (MAE ₹3,676 ± ₹448). The reported test figure of 0.7507 therefore sits at the favourable end of what the model achieves on new data, and the defensible expectation is a range of approximately 0.58 to 0.69 rather than the single-split figure. XGBoost led the tuned cross-validated ranking but was the weakest of the three finalists on test; both are shown so that the ranking is not read off one number alone.

Full metric suite — tuned Random Forest

R^2 0.7507 · MAE ₹3,206 · RMSE ₹5,548 · MAPE 19.67%, against an OLS baseline of R^2 0.5113, MAE ₹3,499 and RMSE ₹7,769. Absolute error correlates with true price at r = 0.818, so error scales with the price level and the model is least reliable at the top of the range.

4.3 Price-tier classification

A secondary price-tier classification model reaches 72.8% accuracy against a 22.3% naive baseline, with a macro F1 of 0.739. Tier boundaries were derived from training data alone, at ₹9,078, ₹13,749 and ₹19,992. The classifier performs cleanly at the bottom of the range (Budget F1 = 0.91) and adequately at the top (Flagship F1 = 0.73), but remains difficult across the middle tiers (Premium F1 = 0.66; Mid F1 = 0.65). This is partly expected where a continuous target is divided into adjacent quartile classes, and the classifier is best used as a rough segmentation signal rather than a substitute for the price model.

4.4 Alternative techniques evaluated

Three further techniques named in the project brief were tested and evaluated on their merits. Two were rejected on evidence and are reported rather than omitted.

Alternative techniques — results and adoption decisions

Technique	Result	Adopted
Feature Selection	Memory, RAM, Front_Camera_MP, Is_5G and Processor_Brand_Unisoc were selected by all three methods. A 10-field "lite" model retains 97.5% of the full model's mean grouped-CV R^2 (0.670 vs. 0.688) at a cost of approximately ₹121 higher mean CV MAE, using 76% fewer inputs. The 0.017 gap is smaller than the full model's own fold-to-fold standard deviation of 0.073. Measured on the test set instead, the lite model retains 96.6% (R^2 0.725 vs. 0.751).	Yes
Dimensionality Reduction (PCA)	Materially reduces accuracy (test R^2 0.751 → 0.519; MAE ₹3,206 → ₹4,256). Random Forests split on thresholds in individual features; rotated components destroy exactly those split points, and 28 of 42 components were needed for 90% variance because one-hot dummies are already near-orthogonal. A genuine negative result, retained in the record.	No
Ensemble Stacking	Marginal held-out gain over the single Random Forest (R^2 0.7517 vs. 0.7507; MAE ₹2,846 vs. ₹3,206). The Ridge meta-learner weights Lasso at 0.46, ahead of Gradient Boosting at 0.32 and the Random Forest at 0.24. The improvement is too small to justify the added complexity and was not independently cross-validated.	No

5. The Central Strategic Insight

The data reveals two distinct pricing regimes, which call for two different playbooks.

– Below the flagship tier — the large majority of the market. Prediction performance is generally stronger and errors are less concentrated. Brand carries little independent pricing power once specifications are fixed, and the top three specifications (Front Camera MP, Battery Level, Memory) account for 68.4% of the model's impurity-based feature importance. Non-flagship brands should compete on spec-per-rupee value.

– At the flagship tier — Apple n = 5, Google n = 9; directional only. A premium is associated with Apple and Google that specifications alone do not explain. Apple averages ₹67,639 and Google ₹43,221 against a dataset mean of ₹16,306, on five and nine listings respectively; within the 128GB+ subset Apple averages 3.47× Samsung and 4.68× Realme on three listings. The model's largest errors concentrate here — it underprices the Galaxy S23 5G by 32.8% and the Motorola Edge 30 Ultra by 42.3%, consistent with these brands shipping flagship-comparable specifications without commanding Apple or Google-level prices. Closing that gap would plausibly require sustained brand-equity investment in marketing, ecosystem and retail experience rather than incremental specification improvements — though this is a hypothesis suggested by the data, not something this dataset can test directly.

6. Business Recommendations

Prioritised recommendations

Priority	Recommendation
High	Price on specifications, not brand alone. Across the feature-importance analyses, Memory, RAM and camera features consistently contributed substantial predictive signal, while brand identity contributed only marginal independent signal once specifications were known.
High	Prioritise accurate Memory and RAM data capture in any pricing tool — these are the two fields with the greatest estimated price association per unit of input.
Medium	Do not treat Brand and Chipset as two independent pricing levers. Their effect sizes are effectively tied ($\varepsilon^2 = 0.366$ vs. 0.368) and their categorical association is strong (Cramér's V = 0.699, on a sparse contingency table and best treated as directional), meaning they carry largely overlapping information; adjusting price for both separately risks double-counting the same signal.
Medium	Treat any "5G adds X% to price" figure as a scenario estimate specific to the device and market segment, not a fixed constant. The ceteris-paribus simulation returned 41.0% on average (range 18.2–64.5%). The raw comparison of REDMI's 5G and non-5G lineups gives +134%, but that conflates 5G with every other specification difference between those lineups — the two numbers should differ, and the gap is itself the finding.
Low	Pilot the lightweight 10-field pricing tool internally, with manual review, for field-sales or retail-partner quoting scenarios where full specification data entry is impractical. It retains 97.5% of the full model's mean CV R^2 at a modest MAE cost.
Low	Treat the tuned Random Forest as a prototype pricing-sanity tool, piloted internally with manual review before any production use. More complex techniques (PCA, stacking) were tested and do not earn their added complexity at this dataset size.

7. Limitations and Caveats

– Sample size ($n = 531$) is modest for statistical inference. Power analysis places the minimum reliably detectable effect at $|r| \approx 0.12$ for correlations and Cohen's $d \approx 0.24$ for two-group comparisons. Anything below those thresholds — and any comparison involving a brand with as few as two listings — should be treated as inconclusive rather than confirmed absent.

– All relationships reported are associations, not proven causal effects. Price is market-set, not derived from a known formula.

– No manufacturing cost, competitor pricing or time-series data was available. The analysis explains which specifications associate with higher prices, not why the market has settled at current price levels.

– Two variables (Battery Level and Screen Size) showed misleading relationships in the full dataset that resolved correctly only after excluding a specific data-quality cluster — a reminder that data-quality issues can silently distort statistical findings if not investigated directly.

– Twelve of the fifteen categorical pairs tested fail the reliability check on expected cell counts. The Brand × Chipset association (Cramér's $V = 0.699$) is bias-corrected and bootstrapped, but the underlying chi-square test remains unreliable, so the figure should be read as directional rather than statistically confirmed.

– Repeated random splits show genuine variance in R^2 (0.584 ± 0.118) around the single reported figure of 0.7507. Selection itself ran on cross-validation alone, but the held-out set was reused for diagnostics after confirmation, so later test-based comparisons are directional.

– Flagged data-quality records (2.8% of listings) are treated as low-confidence predictions and require manual validation before their outputs are used in any pricing decision.

8. Conclusion and Next Steps

The core problem is solved to a defensible standard: price prediction at a held-out test R^2 of 0.7507, cross-validated at a mean of 0.688, using Memory, RAM, camera specifications, Brand and Processor_Brand as the leading drivers, with a secondary price-tier classifier at 72.8% accuracy against a 22.3% naive baseline. Sixty-two figures and eleven strategic questions supporting these conclusions are documented in the companion notebook. Two questions remain genuinely open, and are recorded as open rather than answered speculatively.

Workstream status and recommended next steps

Initiative	Status	Recommendation
Price regression model	Complete	Pilot internally as a pricing-sanity tool with mandatory manual review; a lite 10-feature version is available for field use.
Price-tier classification	Complete	seful as a rough segmentation signal, not a substitute for the price model; Mid, Premium and Flagship remain hard to separate cleanly.
Cost-adjusted margin modelling	Not attempted	Requires manufacturing and bill-of materials cost data not present in this dataset.
Time-series price tracking	Not possible	Requires a listing-date field; open question if that data becomes available.